

# A Corpus-based Approach to Quantifier Scope Disambiguation

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# Introduction

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- Semanticists have generally been concerned with accounting for the range of **possible scopes** for a given sentence.
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## Scope Prediction

- We as semanticists generally do not weigh in on the **actual patterns of usage** of a given possible reading.
- That is, semantics is not concerned with the problem of *quantifier scope disambiguation* (QSD).

# Pragmatics of quantifier scope

In order to develop a model for QSD, we examine the factors influencing quantifier scope in a controlled, but naturally occurring body of real speech: LSAT Logic Puzzles.

## Goal

- Today, our aim is to introduce our corpus and report preliminary findings.

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## Linear order/C-command

(3) Every professor saw a student. *every* >> *a*

(4) A student saw every professor. *a* >> *every*

Gillen 1991, Kutzman & McDonald 1993, Tunstall 1998, Anderson 2004

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(5) Every professor saw a student. *every* >> *a*

(6) A student saw every professor. *a* >> *every*

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**Note:** It is very difficult in English to separate the effect of linear order from the next predictor, grammatical function.

# Psychologically Plausible Predictors

## Grammatical function hierarchy

(7) Joan told a child the story at every intersection. *every* >> *a*

(8) Joan told everyone the story at an intersection. *a* >> *every*

S > Prep > IO > O

Kutzman & McDonald 1993, Tunstall 1998, Micham et al 1980.



# Psychologically Plausible Predictors

## Grammatical function hierarchy

(11) Joan told a child the story at every intersection. *every* >> *a*

(12) Joan told everyone the story at an intersection. *a* >> *every*

$S > \text{Prep} > IO > O$

Kutzman & McDonald 1993, Tunstall 1998, Micham et al 1980.

## loup's (1975) Quantifier Hierarchy

(13) She knows a solution to every problem. *every* >> *a*

(14) She knows a solution to all problems. *a* >> *all*

$\text{each} > \text{every} > \text{all} > \text{most} > \text{many} > \text{several} > \text{some}_{pl} > \text{a few}$

Tunstall 1998, Van Lehn (1978).

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Srinivasan & Yates (2009) show that numerical typicality can be extracted from a large corpus and applied successfully to QSD.

- Applied to a handpicked corpus of 46 items, information about numerical typicality significantly improves prediction, especially for indirect scope.

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## Corpus Worries

- Leave out NPs headed by *a/an*
- Do not separate conjoined or appositive clauses. One result is that the two quantifier do not interact in 61% of the corpus.

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Note: The most active features of each model were things like, intervening comma, intervening conjunct node, intervening quotation mark, intervening colon, etc.

# Summary

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- Linear order/C-command
- Grammatical hierarchy
- Particular quantificational item
- Intervening clause boundaries
- World knowledge



# LSAT Logic Puzzles

## LSATs

- The LSAT exam consists of several types of questions: reading comprehension, analytical reasoning, etc.
- Our corpus is drawn from one particular type of question: Analytical Reasoning Questions, or Logic Puzzles.
- Logic Puzzles follow a particular format as follows:

# Structure of a logic puzzle

In the course of one month Garibaldi has exactly seven different meetings. Each of her meetings is with exactly one of five foreign dignitaries: Fuentes, Matsuba, Rhee, Soleimani, or Tbahi. The following constraints govern Garibaldi's meetings:

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She has exactly three meetings with Fuentes, and exactly one with each of the other dignitaries.

She does not have any meetings in a row with Fuentes.

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2. If Garibaldi's last meeting is with Rhee, then which one of the following could be true?

} Question

- (A) Garibaldi's second meeting is with Soleimani.
- (B) Garibaldi's third meeting is with Matsuba.
- (C) Garibaldi's fourth meeting is with Soleimani.
- (D) Garibaldi's fifth meeting is with Matsuba.
- (E) Garibaldi's sixth meeting is with Soleimani.

} Answers

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- In essence, the entire discourse context is made linguistically explicit, allowing us to extract away from world knowledge.



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## Multiple quantifiers frequent

- Sentences with two or more quantifiers are, unsurprisingly, quite frequent in this register.

# Scopal Domains

## Syntactic Constraints

- In Higgins & Sadock (2003), the sentence was taken as the domain for quantifier scope regardless of syntactic complexity.
- However, it is often clear that a sentence consists of multiple separate scopal domains.
- For example, if two quantifiers appear in a coordinate structure as in (15)

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(17) [Joe ate three oranges]<sub>1</sub> and [Pam ate two apples]<sub>2</sub>.

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# Scopal Domains (cont'd)

- Quotations and Parenthetical content like appositive relative clauses similarly involve multiple scopal domains.
- Therefore, we consider *scopal domains* with multiple quantifiers, rather than *sentences*.
- This is consistent with our stated goal of studying the *pragmatics* of quantifier scope.
- The lack of relative scope between quantifiers in different conjuncts of a coordinate clause is an observation about the syntax/semantics of quantifiers, not their pragmatics.

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- The lack of relative scope between quantifiers in different conjuncts of a coordinate clause is an observation about the syntax/semantics of quantifiers, not their pragmatics.

This approach does not entirely eliminate the role of syntax/semantics in determining possible readings, but it accounts for the most common cases (cf. Higgins & Sadock (2003)'s finding that commas, 'and', etc. are the best predictors in their models)

# Tagging the data

## Procedure

- Quantifier scope by its nature requires a trained linguist to tag.
- First, we separated the data into individual sentences and then further into scopal domains.
- Second, we enlisted undergrads to identify sentences with multiple quantifiers and give a first pass at tagging them.
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No effort was made to quantify inter-annotator agreement since (i) this would require additional skilled coders, and (ii) Higgins & Sadock (2003)'s study did do this, found fairly high variability, and concluded (reasonably) that this is fairly inescapable.



# Categories tagged

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## 3 Factors

- 1 Linear order
- 2 Syntactic position
- 3 Lexical identity of quantifier

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## Lexical identity

- We used a 'splitting' strategy here, having the entire complex determiner tagged in cases like 'more.than.two', 'a.different', etc.

# Examples

## Tagged examples

- (18) Each**&1\_S\_each#** tape is to be assigned to a different**&2\_to\_a.different#** time slot, ...

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- (22) Each**&1\_S\_each#** tape is to be assigned to a different**&2\_to\_a.different#** time slot, ...
- (23) Each**&1\_S\_each#** professor has one or more**&2\_O\_one.or.more#** specialities,
- (24) . . and no**&1\_S\_no#** tape is longer than any other**&2\_than\_any.other#** tape.
- (25) The judge of the show awards exactly four**&1\_O\_exactly.four#** ribbons to four**&1\_to\_four#** of the dogs,

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- 450 sentences, i.e., 900 quantifiers / observations

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S:342, O:199, PREP.MISC:86, A:70, IN:48, P:35 etc.

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## lex.real / lex.real.other

each:170, card.num:158, a:139, exactly:133, no:52, at.least:38 etc.

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  - all show a significant preference for narrow scope relative to S.
  - but they also tend to come last.
  - effects are not significant when controlling for linear order.

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  - all show a significant preference for narrow scope relative to S.
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- For P, and prepositions "on" and PREP.MISC:
  - significant preference for narrow scope relative to S
  - also tend to come last.
  - but effects ARE significant when controlling for linear order.
  - except for P, these effects are no long significant when controlling for the lex.real of the quantifier.

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- we remove the cumulative sentences
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- we focus on S and O only
  - we are left with 489 observations
- we have double counting: some sentences have both an S and an O quantifier and the scope of one completely determines the scope of the other
  - 141 doubly counted sentences
  - we randomly sample one quantifier from each of them



# The S & O dataset without double counting

- 348 observations

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## Response Variable

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## 2 random effects

- lex.real: factor w/ 19 levels "a", "a.different", ...
- lex.real.other: factor w/ 22 levels "a", "a.different", ...

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each:82, a:49, exactly:49, card.num:46, no:33, at.least:17 etc.



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lex.real.other

a:61, each:61, exactly:58, card.num:41, at.least:18, no:16 etc.

# The S & O dataset without double counting

|        | lin.ord: | first |  | lin.ord: | last |
|--------|----------|-------|--|----------|------|
|        | S        | O     |  | S        | O    |
| narrow | 36       | 13    |  | 12       | 76   |
| wide   | 184      | 18    |  | 3        | 6    |

# The S & O dataset without double counting

|        | lin.ord: | first |  | lin.ord: | last |
|--------|----------|-------|--|----------|------|
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## S-wide

- first: **Each chair** is occupied by exactly one representative.
- last: Every week **six crews** – A,B,C,D,E,F– were ranked from first (most productive) to sixth (least productive).

# The S & O dataset without double counting

## S-narrow

- first: **Exactly one child** sits in each chair.
- last: On each day of other days of hiring, **exactly one** worker was hired.

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## S-narrow

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## S-narrow

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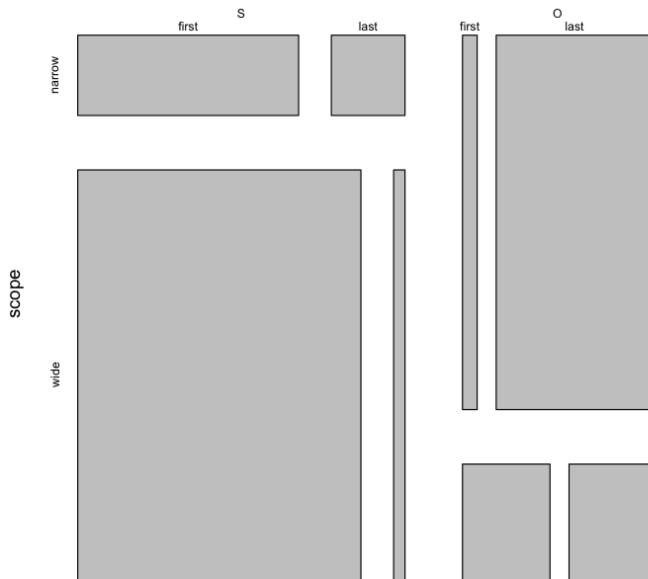
## O-wide

- first: He did not wash **any two of the objects** at the same time.
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## O-narrow

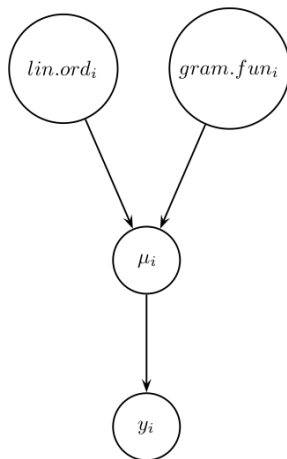
- first: Each mannequin wears exactly **one of the hats**.
- last: Each of his meetings is with **exactly one of the workers**.

# The S & O dataset without double counting



# The fixed-effects logistic regression

$$\text{logit}(\mu_i) = \mu_0 + \beta_{last} \cdot x_{last,i} + \beta_{obj} \cdot x_{obj,i}$$
$$y_i \sim \text{bern}(\mu_i)$$





# The estimated probabilities of / preferences for wide scope

|             | Estimate |
|-------------|----------|
| Intercept   | 1.62     |
| lin.ordlast | -2.93    |
| gram.funO   | -1.27    |

| gram.fun | lin.ord | wide.scope.prob |
|----------|---------|-----------------|
| S        | first   | 0.84            |
| O        | first   | 0.59            |
| S        | last    | 0.21            |
| O        | last    | 0.07            |

# The fixed-effects logistic regression

We show that each of the fixed-effects contributes to the model by performing single term deletions and checking to see that the reduction in deviance is significant.

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|          | Df | Deviance | AIC    |
|----------|----|----------|--------|
| <none>   |    | 296.22   | 302.22 |
| lin.ord  | 1  | 354.80   | 358.80 |
| gram.fun | 1  | 307.77   | 311.77 |

|          | LRT    | Pr(Chi)   |
|----------|--------|-----------|
| lin.ord  | 58.577 | 1.955e-14 |
| gram.fun | 11.547 | 0.0006784 |

# Testing for interactions

We might think that linear order and grammatical function interact in a sort of 'rich get richer' effect. Analysis of deviance shows that this is not the case.

$$M1: \text{logit}(\mu_i) = \mu_0 + \beta_{last} \cdot x_{last,i} + \beta_{obj} \cdot x_{obj,i}$$

$$M2: \text{logit}(\mu_i) = \mu_0 + \beta_{last} \cdot x_{last,i} + \beta_{obj} \cdot x_{obj,i} + \boxed{\beta_{last,obj} \cdot x_{last,i} \cdot x_{obj,i}}$$

|   | Resid. Df | Resid. Dev | Df |
|---|-----------|------------|----|
| 1 |           |            |    |
| 2 | 345       | 296.22     | 1  |

|   | Deviance | P(>  Chi ) |
|---|----------|------------|
| 1 |          |            |
| 2 | 0.031    | 0.86       |

# Model fit

## C

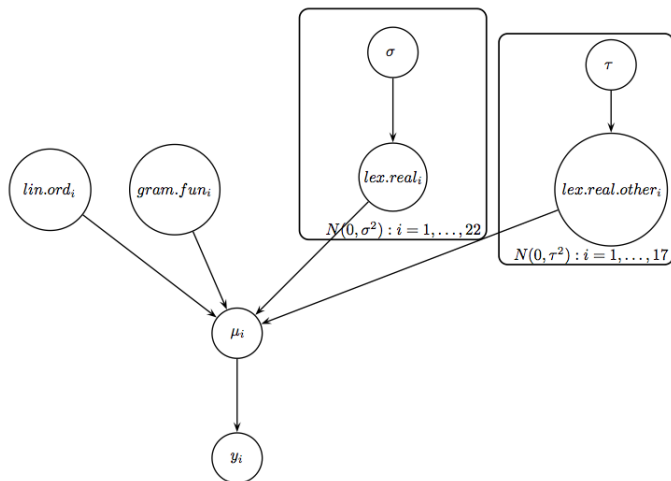
- C is an index of concordance between predicted probability and observed response
- C: 0.83

## Dxy

- Somers' Dxy is a rank correlation between predicted probabilities and observed responses related to C
- Dxy: 0.67

# Adding random intercepts for *lex.real* and *lex.real.other*

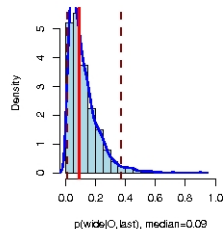
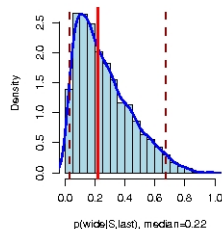
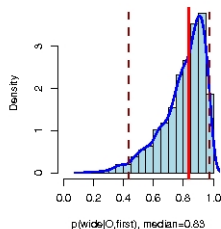
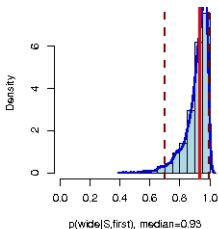
$$\text{logit}(\mu_i) = \mu_0 + \beta_{last} \cdot x_{last,i} + \beta_{obj} \cdot x_{obj,i} + \text{lex.real} + \text{lex.real.other}$$
$$y_i \sim \text{bern}(\mu_i)$$



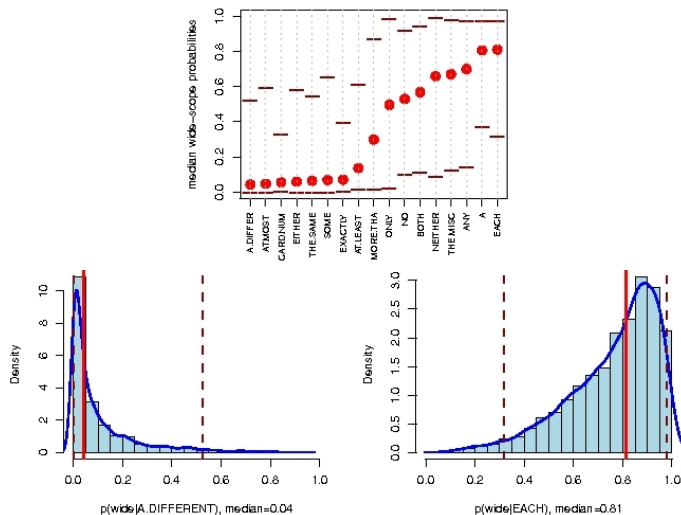
# Adding random intercepts for lex.real and lex.real.other

## Fixed Effects

| Probs |       |     |
|-------|-------|-----|
| S     | first | .93 |
| S     | last  | .22 |
| O     | first | .83 |
| O     | last  | .09 |



# Random effects: lex.real





# Random effects: lex.real

## Decreasing preference for wide scope

each

neither

any

the.miscellaneous

no

both

every

only

# Random effects: lex.real

## Increasing preference for narrow scope

at.least

a.different

some

more.than

exactly

either

at.most

the.same

card.num

# Random effects: lex.real.other

## Decreasing preference for other's wide scope

either

the.same

at.most

modifier.miscellaneous

exactly

more.than

any

a.different

every

both

some

at.least

# Random effects: lex.real.other

## Increasing preference for other's narrow scope

each.other  
the.miscellaneous  
only  
all  
card.num  
neither  
no  
a  
each

# Random-effects logistic regression: model fit

## Mixed effects logistic regression

- C: 0.98
- Dxy: 0.96

# Random-effects logistic regression: model fit

## Mixed effects logistic regression

- C: 0.98
- Dxy: 0.96

## Compare with the fixed-effects logistic regression

- C: 0.83
- Dxy: 0.67

# Random-effects logistic regression: model fit

## Mixed effects logistic regression

- C: 0.98
- Dxy: 0.96

## Compare with the fixed-effects logistic regression

- C: 0.83
- Dxy: 0.67

## Compare with the random-effects regression w/o fixed effects

- C: 0.96
- Dxy: 0.92

## Some Specific Conclusions

Our best model is one with fixed effects for linear order and grammatical function, while modeling the contribution of the actual lexical items as random effects with no interactions.



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Our best model is one with fixed effects for linear order and grammatical function, while modeling the contribution of the actual lexical items as random effects with no interactions.

If asked to choose between various predictors:

- Models with the lexical realization of the quantifiers alone have a better fit than those with fixed effects alone.
- Models with the lexical realization of the other quantifier alone have a better fit than those with a realization of the response quantifier alone.
- Among the fixed effects, linear order is a better predictor than grammatical function (when keeping lex.real/.other random effects, the significance of gram.fun is borderline)

# Acknowledgements

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